

JEE MAIN REVISION MASTER: REDOX REACTIONS

2026 2 QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Redox Reactions — Applied Cases, Reactions & Traps	14 PYQs	Applied Topics
Moderate	Redox Reactions — Standard Calculations & Problem Archetypes	7 PYQs	Quantitative Analysis
Standard	Redox Reactions — Core Principles & Governing Laws	2 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Redox Reactions — Applied Cases, Reactions & Traps	$2+2x=0$, $2[\text{Ag}(\text{CN})_2]^{-}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow [\text{Zn}(\text{CN})_4]^{2-}(\text{aq}) + 2\text{Ag}(\text{s})$, $\text{NH}_4\text{NO}_3, \text{N}$
Redox Reactions — Standard Calculations & Problem Archetypes	Cr^{+3} , I_2 , Cl^{-}
Redox Reactions — Core Principles & Governing Laws	$\text{Cl}_{2(\text{g})} + 2\text{Br}_{(\text{aq})}^{-} \rightarrow 2\text{Cl}_{(\text{aq})}^{-} + \text{Br}_{2(1)}$, $2\text{H}_2\text{O}_{(\text{aq})} \rightarrow 2\text{H}_{2(\text{l})} + \text{O}_{2(\text{g})}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Redox Reactions — Applied Cases, Reactions & Traps (High Yield — 14 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Redox Reactions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $aC_2O_4^{2-} + bMnO_4^- + cH^+ \rightarrow xMn^{2+} + yH_2O + zCO_2$ a and ..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "Given below are two statements. Statement-I: In H_2O_2 , each oxygen atom is assigned an oxidation number of -1, in $...$ " [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $Cl_2(g) + 2Br_{(aq)} \rightarrow ...$ " [Ans: N/A]

3.2 Redox Reactions — Standard Calculations & Problem Archetypes (Moderate Yield — 7 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Redox Reactions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "In the titration of potassium permanganate ($KMnO_4$) against Ferrous ammonium sulphate (FAS) solution, dilute sulphuric acid but not $...$ " [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "The oxidation number of nitrogen atoms in NH_4NO_3 are..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "In which of the following cases a chemical reaction is possible?..." [Ans: N/A]

3.3 Redox Reactions — Core Principles & Governing Laws (Standard Yield — 2 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Redox Reactions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Redox Reactions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Redox Reactions PYQs.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $Cl_2(g) + 2Br_{(aq)} \rightarrow ...$ " [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "Which of the following is not true for oxidation?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$a\text{C}_2\text{O}_4^{2-} + b\text{MnO}_4^- + c\text{H}^+ \rightarrow x\text{Mn}^{2+} + y\text{H}_2\text{O} + z\text{CO}_2$ a and x respectively are

- (A) 5,2 [Correct]
- (B) 4,1
- (C) 3,2
- (D) 4,2

Correct Answer: A

Solution Analysis: In acidic medium, MnO_4^- oxidises oxalate ion $\text{C}_2\text{O}_4^{2-}$ to CO_2 , and itself gets reduced to Mn^{2+} . First write the two half-reactions. For oxalate ion: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2$ Carbon is oxidised here, and the balanced half-reaction is: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$. For permanganate ion in acidic medium: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$. Balancing oxygen and hydrogen using H_2O and H^+ : $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$

Benchmark Problem 2: KCET 2025 (Q#3)

Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{l})$ b. Decomposition reaction ii. $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ c. Displacement reaction iii. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ d. Disproportionation reaction iv. $2\text{H}_2\text{O}_2(\text{l}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$ Choose the correct answer from the options given below.

- (A) a-iv, b-iii, c-i, d-ii
- (B) a-ii, b-i, c-iv, d-iii
- (C) a-iii, b-iv, c-i, d-ii
- (D) a-iii, b-ii, c-i, d-iv

Correct Answer: None

Solution Analysis: Combination reaction - no example given Decomposition reaction - $\text{H}_2\text{O}_2(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ Displacement reaction - $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{l})$ Disproportionation reaction - $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

Benchmark Problem 3: KCET 2020 (Q#1)

The oxidation number of nitrogen atoms in NH_4NO_3 are

- (A) +5,+5
- (B) -3,+5
- (C) +3,-5
- (D) -3,-3

Correct Answer: None

Solution Analysis: $\text{NH}_4\text{NO}_3 \rightarrow \text{NH}_4^+ + \text{NO}_3^-$ For oxidation number of $\text{N}_1 \rightarrow x+4=+1$ $x=+1-4=-3$ So, oxidation state of first nitrogen is -3. For oxidation state of second nitrogen (N_2) $\rightarrow x-6=-1$ $x=-1+6=+5$ So, the answer is -3 and +5.